
What's the current state of local llms less than 50B in size

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BY

disco LATIN • VERB /ˈdis.koː/

“I learn; I become acquainted with.”

from *discere* — to learn

The current state of local LLMs under 50B parameters is marked by a flurry of model releases—including new fine-tunes, architectural variants, and speed-optimized versions—but remains hampered by a near-total lack of published performance or usability benchmarks for local deployment on consumer hardware. No concrete data on tokens per second, memory usage, or task-specific scores appeared in the past week, making it impossible to objectively compare these new models against larger ones or even against earlier sub-50B releases. This gap undermines the practical value of the announcements for developers and users trying to choose a model for local use.

On the hardware front, a notable development was AMD’s June 15 announcement of the Ryzen AI Max+ 395 (“Strix Halo”) mini PC, which claims to load models previously requiring a workstation by offering 128GB of unified memory at under \$2,000. This points toward a possible easing of the memory bottleneck that has constrained local LLM deployment, but the claim has not yet been independently validated with published benchmarks. A second, unnamed tooling or framework update was also noted in the same week, further suggesting industry momentum toward lowering barriers for local LLM use.

Overall, the field is active and optimistic, but the evidence base for actual local performance remains thin. Announced models and hardware outpace the availability of reproducible measurements, leaving the “state of the art” for local sub-50B LLMs defined more by promises than by proven results.

01

Which open-source LLMs under 50B parameters have been released or significantly updated in the past week, including new fine-tunes, quantizations, or architectural changes, and what are their reported capabilities and hardware requirements?

1. Open-Source LLMs Under 50B: Releases and Updates (June 13–20, 2026)

The past week brought several new open-weight models under 50B parameters, spanning fine-tunes, architectural variants, and speed-optimized releases. The sources converge on **GLM-5.2** from Z.ai as the most immediately impactful release for self-hosted use, with both a general-purpose guide and a reference-ranking site naming it the top pick among open-weight models under 50B [\[6\]](#) [\[7\]](#). GLM-5.2 is described as delivering “significant coding + agentic gains” over its predecessor, and it was quickly integrated into Nous Research’s Hermes

02

What concrete performance and usability data (e.g., tokens per second, memory usage, task-specific benchmarks) has been published in the last seven days for local deployment of sub-50B models, and how do these figures compare to state-of-the-art larger models on common consumer hardware?

Based on the provided sources, there is no concrete performance or usability data (such as tokens per second, memory usage, or task-specific benchmarks) for local deployment of sub-50B models published in the last seven days. The available sources are entirely unrelated to the requested topic: one is a social media post about a fintech tutorial using LangChain and StreamLit [1](#), and the third is an Instagram account for an AI newsletter with no substantive content provided.

While two sources offer generalized frameworks or aggregated speed comparisons for LLMs, they do not provide sub-50B specific local deployment figures from the past week. One source describes a benchmarking tool for measuring local LLM performance via Ollama, noting it can compare "results across hardware configurations and model variants," but presents no recent numerical results [2](#). Another source, a speed and latency comparison tool (last updated 2026-06-19), reports aggregate data such as Mercury 2 achieving 789 tok/s and lowest latency figures [3](#), but this data is not filtered by model size (sub-50B) nor tied to local consumer hardware deployment.

The evidence suggests a significant gap in the public availability of fresh, granular performance benchmarks for sub-50B models running locally. The absence of data – despite multiple tools existing to collect it – indicates that either (a) such benchmarks are not being systematically published at a weekly cadence, or (b) the sources available are insufficient to support any comparative analysis between sub-50B and state-of-the-art larger models on common consumer hardware. This report's assessment is that no meaningful synthesis or comparison can be performed from the provided sources alone, and the section topic cannot be addressed with factual grounding.

03

What recent announcements or community developments in the past week concern tooling, frameworks, or commercial products that specifically target running LLMs under 50B on local devices (e.g., updated llama.cpp support, new edge AI hardware, or local-first applications)?

The past week produced two convergent developments aimed at lowering the barriers to running LLMs on personal devices. On June 15, AMD announced the Ryzen AI Max+ 395 (“Strix Halo”) mini PC, claiming it can load models that previously required a workstation by providing 128GB of unified memory at under \$2,000 [\[4\]](#). On June 20, the open-source workspace Odysseus—created by the most-subscribed independent YouTube creator—was reported to have passed 74,000 stars, offering a self-hosted environment that runs LLMs on user-owned hardware [\[4\]](#). Together, these two announcements address complementary bottlenecks: one reduces hardware cost and memory constraints; the other lowers software friction for local deployment.

AMD’s Strix Halo is a vendor claim that must be weighed against its technical trade-offs. The company states that the mini PC’s 128GB of unified memory allows loading models up to ~200 billion parameters, far exceeding the 24–32GB available on consumer GPUs like the RTX 4090/5090 [\[5\]](#). However, independent testing on Linux shows that only about 108GB is exposed to the GPU, and the system’s 256 GB/s memory bandwidth trails Apple’s M-series (546–819 GB/s) [\[5\]](#). For models under 50B parameters—the focus of this section—the bandwidth limitation is less severe, but it still means that even moderately sized models will generate tokens more slowly than on Apple Silicon. The cost advantage (\$2,000 for 128GB) is notable, but the performance gap suggests that Strix Halo is best suited for batch or offline inference rather than interactive use.

Odysseus represents a community-driven tooling development with broad reach: the blog notes it is free and self-hosted, and its rapid adoption (74k+ stars) signals strong grassroots interest in local inference [\[4\]](#). Unlike AMD’s hardware announcement, Odysseus is a software project that can run on existing hardware, including Apple Silicon and presumably Strix Halo once available. The project’s provenance—from a creator with a massive audience—likely accelerated its visibility, but the underlying pattern of self-hosted workspaces is not new. What matters is that Odysseus lowers the software friction for users who want to run LLMs locally without cloud dependencies.

Taken together, these two developments illustrate a maturing ecosystem where both hardware and software are evolving to support local LLM deployment. The hardware side (AMD) offers a lower-cost entry point for memory-intensive models, but with a bandwidth penalty that limits real-time performance. The software side (Odysseus) provides a polished, free interface that can leverage whatever hardware is available. The evidence suggests that the primary barrier for models under 50B is no longer whether they can run locally, but how well they run

—and the past week’s announcements address that question from opposite ends. A comparison of memory bandwidth across platforms highlights the

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- [2] [LLM Benchmark - Local LLM Performance Testing](https://llm.aidatools.com/) — <https://llm.aidatools.com/>
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- [3] [LLM Speed & Latency Comparison — Tokens/sec & Response Latency \(2026\) | BenchLM.ai](https://benchlm.ai/llm-speed) — <https://benchlm.ai/llm-speed>
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- [7] [Best Open Source LLMs | LLM Reference](https://www.llmreference.com/best/open-source) — <https://www.llmreference.com/best/open-source>
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